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PIERRE AZOULAY

employment

Massachusetts Institute of Technology, Sloan School of Management, 2006-present.
International Programs Professor of Management, 2016-present.
Associate Professor of Strategy, 2009-2015.
Assistant Professor of Strategy, 2006-2009.

Columbia University, Graduate School of Business, 2001-2006.
Associate Professor of Management, 2005-2006.
Assistant Professor of Management, 2001-2005.

other affiliations

National Bureau of Economic Research, 2003-present.
Research Associate, Productivity, Innovation, & Entrepreneurship Program, 2011-present.
Faculty Research Fellow, Productivity Program, 2003-2011.

Institute for Progress, Washington, DC, 2022-present.
Senior Fellow.

education

Ph.D., Management, Massachusetts Institute of Technology, 2001.
M.A., Telecommunications, Michigan State University, 1994.
Diplôme d'Études Supérieures de Gestion, Institut National des Télécommunications, 1994.
Classes Préparatoires Scientifiques, Lycée Saint Louis, 1988-1991.

publications

Working papers

Pierre Azoulay, Raffaella Sadun, and Daniela Scur. "Before the Exodus? Young Scientists and the Future of US Science." NBER Working Paper #35330.

Pierre Azoulay and Wesley H. Greenblatt. "Does Grant Peer Review Penalize Scientific Risk Taking? Evidence from the NIH." NBER Working Paper #33495. *Resubmitted.*

Journal articles

Pierre Azoulay, Daniel P. Gross, and Bhaven N. Sampat. 2026. "Indirect Cost Recovery in U.S. Innovation Policy: History, Evidence, and Avenues for Reform." *Entrepreneurship & Innovation Policy in the Economy* **5**: 133-182.

Pierre Azoulay, Matthew Clancy, Danielle Li, and Bhaven N. Sampat. 2025. "What if NIH had been 40% smaller?" *Science* **389**(6767): 1303-1305.

Pierre Azoulay, Joshua L. Krieger, and Abhishek Nagaraj. 2025. "Old Moats for New Models: Openness, Control, and Competition in Generative AI." *Entrepreneurship & Innovation Policy and the Economy* **4**: 7-46.

Shumin Qiu, Claudia Steinwender, and Pierre Azoulay. 2025. "Paper Tiger? Chinese Science and Citation Home-bias." *Journal of International Economics* **157**: 104123.

Pierre Azoulay, Misty Heggeness, and Jennifer Kao. 2025. "Medical Progress and Health Care Finance: Evidence from Academic Medical Centers." *Research Policy* **54**(2): 105163.

Shumin Qiu, Claudia Steinwender, and Pierre Azoulay. 2025. "Who Stands on the Shoulders of Chinese (Scientific) Giants? Evidence from Chemistry." *Research Policy* **54**(1): 105147.

Pierre Azoulay, Benjamin Jones, J. Daniel Kim, and Javier Miranda. 2022. "Immigration and Entrepreneurship in the United States." *American Economic Review: Insights* **4**(1): 71-88.

Pierre Azoulay, Wesley H. Greenblatt, and Misty L. Heggeness. 2021. "Long-term Effects from Early Exposure to Research: Evidence from the NIH 'Yellow Berets.' " *Research Policy* **50**(9): 104332.

Pierre Azoulay and Freda B. Lynn. 2020. "Self-Citation, Cumulative Advantage, and Gender Inequality in Science." *Sociological Science* **7**: 152-186.

Joshua Angrist, Pierre Azoulay, Glenn Ellison, Ryan Reed Hill, and Susan Feng Lu. 2020. "Inside Job or Deep Impact? Using Extramural Citations to Assess Economic Scholarship." *Journal of Economic Literature* **58**(1): 3-52.

Pierre Azoulay, Benjamin Jones, J. Daniel Kim, and Javier Miranda. 2020. "Age and High-Growth Entrepreneurship." *American Economic Review: Insights* **2**(1): 65-82.

Pierre Azoulay, J. Michael Whalen, and Ezra Zuckerman. 2019. "Death of the Salesman, but not the Sales Force: How Interested Promotion Skews Scientific Valuation." *American Journal of Sociology* **125**(3): 786-845.

Pierre Azoulay, Christian Fons-Rosen, and Joshua S. Graff Zivin. 2019. "Does Science Advance One Funeral at a Time?" *American Economic Review* **109**(8): 2889-2920.

Pierre Azoulay, Erica Fuchs, Anna P. Goldstein, and Michael Kearney. 2019. "Funding Breakthrough Research: Promises and Challenges of the 'ARPA Model.' " *Innovation Policy and the Economy* **19**: 69-96.

Pierre Azoulay, Joshua Graff Zivin, Danielle Li, and Bhaven Sampat. 2019. "Public R&D Investments and Private Sector Patenting: Evidence from NIH Funding Rules." *Review of Economic Studies* **86**(1): 117-152.

Pierre Azoulay, Brian Reschke, and Toby E. Stuart. 2018. "Status Spillovers: The Effect of Status-Conferring Prizes on the Allocation of Attention." *Administrative Science Quarterly* **63**(4): 819-847.

Pierre Azoulay, Alessandro Bonatti, and Joshua L. Krieger. 2017. "The Career Effects of Scandal: Evidence from Scientific Retractions." *Research Policy* **46**(9): 1552-1569.

Joshua Angrist, Pierre Azoulay, Glenn Ellison, Ryan Hill, and Susan Lu. 2017. "Economic Research Evolves: Fields and Styles." *AER: Papers & Proceedings* **107**(5): 293-297.

Pierre Azoulay, Danielle Li, and Bhaven N. Sampat. 2017. "The Applied Value of Public Investments in Biomedical Research." *Science* **356**(6333): 78-81.

Pierre Azoulay, Ina Ganguli, and Joshua Graff Zivin. 2017. "The Mobility of Elite Life Scientists: Professional and Personal Determinants." *Research Policy* **46**(3): 573-590.

Pierre Azoulay, Christopher Liu, and Toby Stuart. 2017. "Social Influence Given (Partially) Deliberate Matching: Career Imprints in the Creation of Academic Entrepreneurs." *American Journal of Sociology* **122**(4): 1223-1271.

Pierre Azoulay, Jeffrey L. Furman, Joshua L. Krieger, and Fiona E. Murray. 2015. "Retractions." *Review of Economics and Statistics* **97**(5): 1118-1136.

Pierre Azoulay, Toby Stuart, and Yanbo Wang. 2014. "Matthew: Effect or Fable?" *Management Science* **60**(1): 92-109.

Pierre Azoulay, Joshua Graff Zivin, and Gustavo Manso. 2013. "NIH Peer Review: Challenges and Avenues for Reform." *Innovation Policy and the Economy* **13**: 1-21.

Pierre Azoulay. 2012. "Empirical Studies of the Diffusion of Scientific and Technological Knowledge: Some Methodological Considerations." *Economia e Politica Industriale—Journal of Industrial and Business Economics* **39**(3): 73-87.

Pierre Azoulay, Gustavo Manso, and Joshua Graff Zivin. 2011. "Incentives and Creativity: Evidence from the Academic Life Sciences." *RAND Journal of Economics* **42**(3): 527-554.

Pierre Azoulay, Nelson P. Repenning, and Ezra W. Zuckerman. 2010. "Nasty, Brutish, and Short: Embeddedness Failure in the Pharmaceutical Industry." *Administrative Science Quarterly* **55**(3): 472-507.

Pierre Azoulay, Joshua Graff Zivin, and Jialan Wang. 2010. "Superstar Extinction." *Quarterly Journal of Economics* **125**(2): 549-589.

Pierre Azoulay, Waverly Ding, and Toby Stuart. 2009. "The Effect of Academic Patenting on the Rate, Quality, and Direction of (Public) Research Output." *Journal of Industrial Economics* **57**(4): 637-676.

Pierre Azoulay, Bhaven Sampat, and Ryan Michigan. 2007. "The Anatomy of Medical School Patenting." *New England Journal of Medicine* **357**(20): 2049-2056.

Pierre Azoulay, Waverly Ding, and Toby Stuart. 2007. "The Determinants of Faculty Patenting Behavior: Demographics or Opportunities?" *Journal of Economic Behavior & Organizations* **63**(4): 599-623.

Pierre Azoulay, Andrew Stellman, and Joshua Graff Zivin. 2006. "PublicationHarvester: An Open-Source Software Tool for Science Policy Research." *Research Policy* **35**(7): 970-974.

Pierre Azoulay. 2004. "Capturing Knowledge Within and Across Firm Boundaries: Evidence from Clinical Development." *American Economic Review* **94**(5): 1591-1612.

Pierre Azoulay, Ernst Berndt, and Robert Pindyck. 2003. "Consumption Externalities and Diffusion in Pharmaceutical Markets: Anti-Ulcer Drugs." *Journal of Industrial Economics* **51**(2): 243-270.

Pierre Azoulay. 2002. "Do Pharmaceutical Sales Respond to Scientific Evidence?" *Journal of Economics and Management Strategy* **11**(4): 551-594.

Pierre Azoulay and Scott Shane. 2001. "Entrepreneurs, Contracts, and the Failure of Young Firms." *Management Science* **47**(3): 337-358.

Chapters in edited volumes and other non-peer-reviewed materials

Pierre Azoulay, Daniel P. Gross, Lisa Larrimore Ouellette, and Bhaven Sampat. 2025. "Brief of Scholars of Economics and Innovation as Amici Curiae in Support of Plaintiffs-Appellees in *Massachusetts v. NIH*." Nos. 25-1343, 25-1344, 25-1345 (1st Cir. 2025).

Pierre Azoulay and Danielle Li. 2022. "Scientific Grant Funding." Chapter 4 in *Innovation and Public Policy*, edited by Austan Goolsbee and Benjamin F. Jones, 117-150. Chicago: University of Chicago Press.

Pierre Azoulay, Anna Goldstein, Joshua Graff Zivin, and Vladimir Bulovic. 2017. "Promoting Energy Innovation with Lessons from Drug Development." Hamilton Project Policy Proposal #2017-16.

Pierre Azoulay, Joshua Graff Zivin, and Bhaven Sampat. 2012. "The Diffusion of Scientific Knowledge Across Time and Space: Evidence from Professional Transitions for the Superstars of Medicine." Chapter 2 in *The Rate & Direction of Inventive Activity Revisited*, edited by Joshua Lerner and Scott Stern, 107-155. Chicago: University of Chicago Press.

Pierre Azoulay and Joshua Lerner. 2012. "Technological Innovation and Organizations." Chapter 14 in *The Handbook of Organizational Economics*, edited by Robert Gibbons and John Roberts, 575-603. Princeton, NJ: Princeton University Press.

Zombie papers

Pierre Azoulay and Ariel Fishman. 2020. "The Rise of For-Profit Experimental Medicine." NBER Working Paper #26892.

Punditry

Pierre Azoulay. 2026. "In (Qualified) Defense of the Research Project Grant." *Science* **392**(6803): 1101.

Pierre Azoulay. 2012. "Turn the Scientific Method on Ourselves." *Nature* **484**(7392): 31-32.

Pierre Azoulay. 2019. "Small-team Science is Beautiful." *Nature* **566**(7744): 330-332.

Pierre Azoulay and Benjamin Jones. 2020. "Beat COVID-19 Through Innovation." *Science* **368**(6491): 553.

grants & awards

National Institutes of Health (co-PI), $4.362M	2013-2018
National Science Foundation (co-PI), $428K	2017-2019
National Science Foundation (co-PI), $376K	2015-2017
National Science Foundation (co-PI), $347K	2011-2014
National Science Foundation (co-PI), $250K	2007-2009
International Network on the Value of Health Research (NIH), $20K	2017-2018
Kaufman Foundation/NBER IPE Program, $20K	2007-2008
Alfred P. Sloan Industry Studies Fellow, $40K	2005-2007
The Merck Foundation (co-investigator), $600K	2003-2007
Institute for Social & Economic Research & Policy (co-investigator), $9K	2002
Columbia Alliance for Health Care Management (co-investigator), $13K	2002
Fellowship, MIT	2000
Fellowship, Crédit Suisse Holding, MIT	1998
Prix d'Excellence, Informatique CDC, Paris	1992

service

Group Head, Technological Innovation, Entrepreneurship, and Strategy, MIT Sloan, 2016-2019, 2020-2026.

Member of the Editorial Board, *The Review of Economics and Statistics*, 2021-2026.

Associate Editor, *Management Science*, 2011-2019.

PhD Program Coordinator, Technological Innovation, Entrepreneurship, and Strategy, MIT Sloan, 2011-2019.

Committee on Evaluation of the Advanced Research Projects Agency—Energy, The National Academies, 2015-2017.

Organizer, Summer Institute Meetings, NBER Program on Innovation Policy & the Economy, 2007, 2010-2013.

Organizer, Fall 2005 Meetings, NBER Program on Technological Progress and Productivity Measurement.

Program Committee, International Industrial Organization Conference, 2008-2009.

MIT Undergraduate Practice Opportunities Program (UPOP), Strategy Module, IAP 2007-2008.

Ad-hoc reviewer for *Management Science*, *Science*, *Research Policy*, *Economics of Innovation and New Technology*, *The Review of Economics and Statistics*, *Administrative Science Quarterly*, *The Quarterly Journal of Economics*, *The American Economic Review*, *The Review of Economic Studies*, *Econometrica*, *The Journal of Political Economy*, *The Review of Industrial Organization*, and the National Science Foundation.

teaching

MBA: Competitive Strategy; Innovation Strategy; Entrepreneurial Strategy; Organizing for Innovation.
Executive MBA: Competitive Strategy; Platform Strategy.
Sloan Fellows: Innovation-driven Advantage.
PhD: Economics of Ideas & Innovation.

thesis supervision

Student	*Program*	*Initial Placement*
Ariel Fishman	PhD, Columbia University	Yeshiva University
Suchin Virabhak	PhD, Columbia University	Pfizer Corp.
Jialan Wang	PhD, MIT	Wash. University in St. Louis
Christopher Liu	DBA, Harvard Business School	University of Toronto
Patrick Gaulé	PhD, EPFL	CERGE-EI
Ina Ganguli	PhD, Harvard Kennedy School	Stockholm School of Economics
Danielle Li	PhD, MIT Economics	Northwestern University
Abhishek Nagaraj	PhD, MIT Sloan	UC Berkeley
Joshua Krieger	PhD, MIT Sloan	Harvard Business School
Jorge Guzman	PhD, MIT Sloan	Columbia University
J. Daniel Kim	PhD, MIT Sloan	University of Pennsylvania
Jennifer Kao	PhD, Harvard Kennedy School	UCLA
Samantha Zyontz	PhD, MIT Sloan	Stanford Law School (postdoc)
Hyejun Kim	PhD, MIT Sloan	HEC Paris
Ankur Chavda	PhD, MIT Sloan	HEC Paris
Caroline Fry	PhD, MIT Sloan	University of Hawaii
Ryan Hill	PhD, MIT Economics	Brigham Young University
Carolyn Stein	PhD, MIT Economics	UC Berkeley
Jane Wu	PhD, MIT Sloan	UCLA
Soomi Kim	PhD, MIT Sloan	Columbia University
Wesley Greenblatt	PhD, MIT Sloan	Harvard Medical School
Luca Gius	PhD, MIT Sloan	Bocconi University
Christina Nguyen	PhD, MIT Sloan	Fordham University

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